

## 4. Belle River

Shoreline length: 63 km



### Shore Types

Belle River is an exposed stretch of diverse coast characterized mainly by wetlands and low plains along the south-east end of PEI.

% Shore types within SMU

| low plain | bluff | sloped armour | cliff | wetland | vertical struct. | dunes | sandspit |
|-----------|-------|---------------|-------|---------|------------------|-------|----------|
| 25        | 7     | 2             | 10    | 40      | 0                | 10    | 6        |



Dunes, sandspit and estuarine wetlands north of Big Point

### Governance, Infrastructure, and Economy

This SMU is almost completely located within the Rural Municipality of Belfast (~94%).

Key infrastructure includes a portion of the Trans-Canada Highway and provincial roads: Ponds Road to Ponds Road Beach (South Pinette); Gascoigne Road (South Pinette); the TCH at Bears Way, Flat River, and Flat River to Big Point Road; Camp Road (Route 261, upper Flat River); Big Point Road at the TCH to its end; the TCH from Flat River to Belle River; Culloden Road (Route 203, Flat River at the TCH); Nicholson Road (Belle River, to the beach); Stewart Point Road (Belle River, at the coast); and Factory Road (Belle River). Coastal economic drivers are limited. Local and legacy land use constraints include unpaved coastal roads running parallel to the coast.

### Natural Assets and Heritage

There are no Provincial or Federal Parks nor Wildlife Management Areas in Belle River, however Ponds Road Beach is located in this SMU.

The shoreline includes critical habitat for bird species including the endangered Piping Plover and threatened Bank Swallow.

## Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

## Sediment transport

The estimated net sediment transport rate averaged across the SMU is high and feeds numerous beaches and sandspits. Sediment transport rate potential is highest along the center of the SMU.

## Coastal Flood and Erosion Risk

The area has relatively low built up % though some infrastructure exists close to the shore. Due to the primary shore types and exposure to waves the area experiences moderate flood and erosion hazards. The low plains, sandspits, and wetlands are vulnerable to increasing flood hazards. Erosion hazard is highest along the exposed dune.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

|           | FLOOD RISK         |                                  |                                   |                                 | EROSION RISK          |                        |                      |
|-----------|--------------------|----------------------------------|-----------------------------------|---------------------------------|-----------------------|------------------------|----------------------|
|           | Present            | Short-term<br>0.3m SLR<br>(2050) | Medium-term<br>0.6m SLR<br>(2080) | Long-term<br>1.2m SLR<br>(2130) | Short-term<br>(~2050) | Medium-term<br>(~2080) | Long-term<br>(~2130) |
| Buildings | 2 [<1/km]          | 3 [<1/km]                        | 4 [<1/km]                         | 12 [<1/km]                      | 2 [<1/km]             | 9 [<1/km]              | 10 [<1/km]           |
| Roads     | 996 m [16<br>m/km] | 1332 m [21<br>m/km]              | 1806 m [28<br>m/km]               | 3025 m [48<br>m/km]             | 100 m [2 m/km]        | 242 m [4 m/km]         | 542 m [9 m/km]       |

## Proposed SMP Strategies

|                         | Short-term                                                                                                                                                                                                                                          | Medium-term | Long-term   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| Sea level rise range    | 0 – 0.3 m                                                                                                                                                                                                                                           | 0.3 – 0.6 m | 0.6 – 1.2 m |
| Potential timeframe     | 2030 - 2050                                                                                                                                                                                                                                         | 2050 - 2080 | 2080 - 2130 |
| <b>4.00 Belle River</b> | <b>NAI</b>                                                                                                                                                                                                                                          | <b>MR</b>   | <b>MR</b>   |
| Rationale               | No Active Intervention is proposed for the short-term, followed by Managed Realignment. This will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including dunes, beaches, wetlands and cliff habitat. |             |             |

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.